Symmetry in Exact diagonalization method

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The central problem of many-body physics is to diagonalize the Hamiltonian $\mathcal{H}$ in the Hilbert space $\mathscr{H}$. To be explicit, in computational basis we have

$$\mathscr{H} = \mathrm{span}\{|\sigma_1 \dots \sigma_L\rangle \mid \sigma_i = 1, \dots, d\}. \tag{1}$$

The problem with a naive approach is that $\mathscr{H}$ is exponentially large. Let L be the lattice size and d be the local Hilbert space dimension, the dimension of $\mathscr{H}$ is d^L. If the system has $\mathsf{U}(1)$ symmetry, then $\mathscr{H}$ is further decomposed into subspaces $\mathscr{H}_N$ with fixed particle number N, defined as

$$\mathscr{H}_N = \mathrm{span}\{|\sigma_1 \dots \sigma_L\rangle \mid \sum_{i=1}^{L} \sigma_i = N\}, \tag{2}$$

and the dimension of $\mathscr{H}_N$ is $\binom{L}{N}$, which is still exponentially large.

To address such problem, we will use the representation theory[1] to decompose the Hilbert space into irreducible representations of the symmetry group G of the Hamiltonian $\mathcal{H}$.

We will assume G is a finite group, and it acts on the Hilbert space $\mathscr{H}$ via a unitary representation $U : G \to U(\mathscr{H})$. What we mean by $\mathcal{H}$ has symmetry G is that $\mathcal{H}$ commutes with the action of G, i.e. $[U(g), \mathcal{H}] = 0$ for all $g \in G$. By Schur's lemma, we know that $\mathcal{H}$ is block diagonal in the basis of irreducible representations of G. Hence we can diagonalize $\mathcal{H}$ in each irreducible representation separately, which reduces the computational cost significantly.

1 Building the symmetry group

In our case case, we consider two kinds of symmetry: the lattice symmetry that permutes the sites and the internal symmetry that performs a spin flip:

$$g\left|\sigma_1 \dots \sigma_L\right\rangle = \left|\sigma_{g^{-1}(1)} \dots \sigma_{g^{-1}(L)}\right\rangle, \quad g \in G_{\text{latt}} \tag{3}$$
$$\mathcal{F}\left|\sigma_1 \dots \sigma_L\right\rangle = \left|(d-\sigma_1), \dots, (d-\sigma_L)\right\rangle \tag{4}$$

and the full symmetry group is $G = G_{\text{latt}} \times \mathbb{Z}_2$. The lattice symmetry group G_{latt} is further decomposed into a semi-direct product of the translation group G_{tr} and the point group G_{pt}, i.e. $G_{\text{latt}} = G_{\text{tr}} \rtimes G_{\text{pt}}$, where $G_{\text{tr}} = \mathbb{Z}_{N_1} \times \cdots \mathbb{Z}_{N_d}$ is a finite Abelian group and G_{pt} is a finite group.

Suppose we already know all the information of G_{pt}, we need to Calculate the character table of G_{latt} and G.

The lattice Group First we describe the structure of the lattice group $G_{\text{latt}} = G_{\text{tr}} \rtimes G_{\text{pt}}$. Let $\mathbf{t} \in G_{\text{tr}}$ and $r \in G_{\text{pt}}$. The element in G_{latt} can be written as $(\mathbf{t}, r)$. It acts on the real space as

$$(\mathbf{t}, r)\mathbf{x} = r\mathbf{x} + \mathbf{t}. \tag{5}$$

The group multiplication defined as

$$(\mathbf{t}_1, r_1)(\mathbf{t}_2, r_2) = (\mathbf{t}_1 + r_1\mathbf{t}_2, r_1 r_2) \tag{6}$$

where $r(\mathbf{t})$ is the action of r on $\mathbf{t}$. The inverse of $(\mathbf{t}, r)$ is given by

$$(\mathbf{t}, r)^{-1} = (-r^{-1}\mathbf{t}, r^{-1}). \tag{7}$$

The irreducible representations of G_{tr} are one-dimensional, and are labeled by the momentum

$$\mathbf{k} \in \text{BZ} = \left\{(k_1, \dots, k_d) \mid k_i = 0, \frac{2\pi}{N_i}, \dots, \frac{2\pi(N_i - 1)}{N_i}\right\}, \tag{8}$$

with the character defined as

$$\chi_{\mathbf{k}}^{G_{\text{tr}}}(\mathbf{t}) = \mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\mathbf{t}}. \tag{9}$$

The irreducible representations of G_{latt} are labeled by the momentum $\mathbf{k}$ and the irreducible representations of the little group

$$G_{\mathbf{k}} = \{r \in G_{\text{pt}} \mid r\mathbf{k} = \mathbf{k}\} \subset G_{\text{pt}}. \tag{10}$$

Suppose we know all the irreducible representations of the little group $G_{\mathbf{k}}$, with its character denoted as $\chi_\alpha^{G_{\mathbf{k}}}$, then the irreducible representations of G_{latt} are given by

$$\chi_{\mathbf{k},\alpha}^{G_{\text{latt}}}(t,r) = \sum_{g \in G_{\text{pt}}/G_{\mathbf{k}}} \chi_{\mathbf{k}}^{G_{\text{tr}}}(gt)\chi_\alpha^{G_{\mathbf{k}}}(grg^{-1}) \tag{11}$$

where g runs over the left cosets of $G_{\mathbf{k}}$ in G_{pt}.

Full Symmetry Group The full symmetry group is $G = G_{\text{latt}} \times \mathbb{Z}_2$. The character of an element $(\mathbf{t}, r, f)$ is given by

$$\chi_{\mathbf{k},\alpha,f}^{G}(t,r,s) = \chi_{\mathbf{k},\alpha}^{G_{\text{latt}}}(t,r)\chi_f^{\mathbb{Z}_2}(s) \tag{12}$$

where $f = \pm 1$ labels the two irreducible representations of $\mathbb{Z}_2$. We will use the character (12) to decompose the Hilbert space into irreducible representations of G.

2 Projectors and Adapted Basis

Let $i = (\mathbf{k}, \alpha, f)$ label the irreducible representations of G. The projector onto the irreducible representation i is given by

$$\Pi_i = \frac{d_i}{|G|} \sum_{g \in G} \chi_i(g) g, \tag{13}$$

where d_i is the dimension of the irreducible representation. The adapted basis of such irreducible representation is given by

$$\frac{1}{N_i^x} \Pi_i \left|x\right\rangle = \frac{1}{N_i^x} \frac{d_i}{|G|} \sum_{g \in G} \chi_i(g) g \left|x\right\rangle, \quad x \in \mathscr{H}. \tag{14}$$

where N_i^x is the normalization factor. $|x\rangle$ is chosen as the representative of the orbit of G in $\mathscr{H}$. For a deterministic output, we choose the representative $|x\rangle$ to be the one with the smallest integer value when we map $|\sigma_1 \dots \sigma_L\rangle$ to an integer by

$$|\sigma_1 \dots \sigma_L\rangle \mapsto \sum_{i=1}^{L} \sigma_i d^{i-1}. \tag{15}$$

The normalization factor N_i^x is given by

$$(N_i^x)^2 = \langle x|\Pi_i|x\rangle = \frac{d_i}{|G|}\sum_{g\in G}\chi_i(g)\,\langle x|g|x\rangle\,. \tag{16}$$

that is, we only need to sum over the stabilizer

$$\mathrm{Stab}_G(|x\rangle) = \{g\in G \mid g\,|x\rangle = |x\rangle\} \tag{17}$$

and the normalization factor is given by

$$(N_i^x)^2 = \frac{d_i}{|G|}\sum_{g\in \mathrm{Stab}_G(|x\rangle)}\chi_i(g). \tag{18}$$

3 Hamiltonian in Adapted Basis

Let $|i,x\rangle = \frac{1}{N_i^x}\Pi_i\,|x\rangle$ be the adapted basis of the irreducible representation i. Since we will dealing with invariant Hamiltonian, that is,

$$\frac{1}{N}\sum_{g\in G} g\mathcal{H}_{\mathrm{loc}}g^{-1} \tag{19}$$

where N is the number of stabilizers of $\mathcal{H}_{\mathrm{loc}}$ under conjugate action of G. We have

$$\langle i,x|\mathcal{H}|i,x'\rangle = \frac{1}{N_i^x N_i^{x'}}\,\langle x|\Pi_i\mathcal{H}\Pi_i|x'\rangle \tag{20}$$

$$= \frac{1}{N_i^x N_i^{x'}}\frac{d_i^2}{|G|^2}\sum_{g,g'\in G}\frac{1}{N}\sum_h \chi_i(g)\chi_i(g')\,\langle x|gh\mathcal{H}_{\mathrm{loc}}h^{-1}g'^{-1}|x'\rangle \tag{21}$$

$$= \frac{1}{N_i^x N_i^{x'}}\frac{d_i^2}{|G|^2}\sum_{g,g'\in G}\frac{1}{N}\sum_h \chi_i(gh^{-1})\chi_i(g'h^{-1})\,\langle x|g\mathcal{H}_{\mathrm{loc}}g'^{-1}|x'\rangle \tag{22}$$

References

[1] William Fulton and Joe Harris. *Representation Theory: A First Course*. Springer, 1991.